

Machine Learning for Images

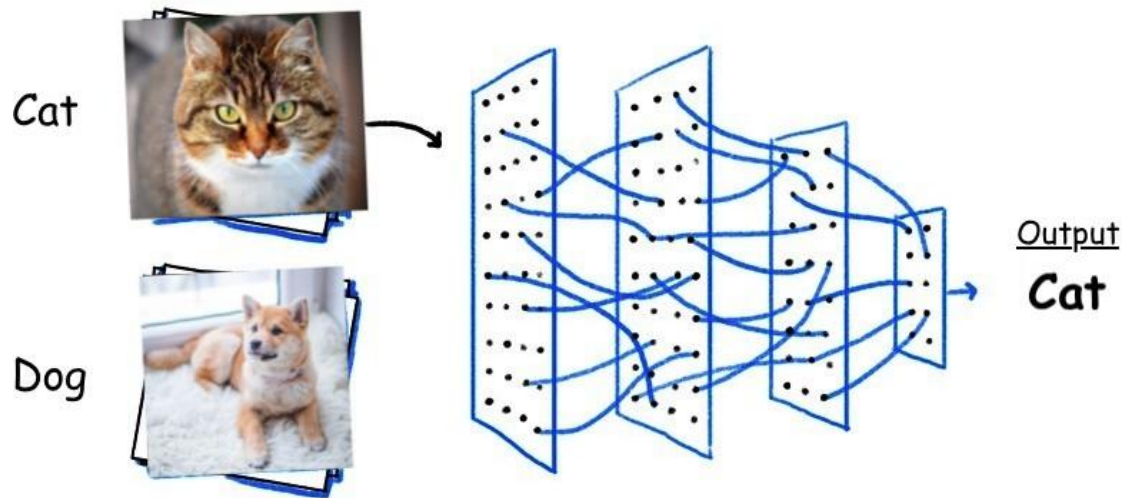


Image Classification

- **What?**
 - Assigning labels to an entire image.
- **Where:**
 - Identifying objects in photos (e.g., cats, dogs, cars)
 - Diagnosing medical images (e.g., detecting tumors in X-rays or MRIs).
- **How?**
 - Convolutional Neural Networks (CNNs).
 - Pre-trained models like ResNet, VGG, and Inception.

Image Classification

- Image classification is a supervised learning problem: **define** a set of **target classes** (objects to identify in images), and train a model to **recognize them using labeled example photos**.



Unsupervised Learning

- Unsupervised learning technique is a fully automated method that **does not leverage training data**.
- This means machine learning algorithms are used to **analyze and cluster unlabeled datasets by discovering hidden patterns or data groups** without the need for human intervention.
- With the help of a suitable algorithm, the particular **characterizations of an image are recognized** systematically during the image processing stage.

Supervised Classification

- Supervised image classification methods use **previously classified reference samples** (the ground truth) in order to **train the classifier** and subsequently classify new, unknown data.
- Therefore, the supervised classification technique is the process of **visually choosing samples of training data within the image and allocating them to pre-chosen categories**, including vegetation, roads, water resources, and buildings.
- This is done to create statistical measures to be applied to the overall image.

Supervised Classification

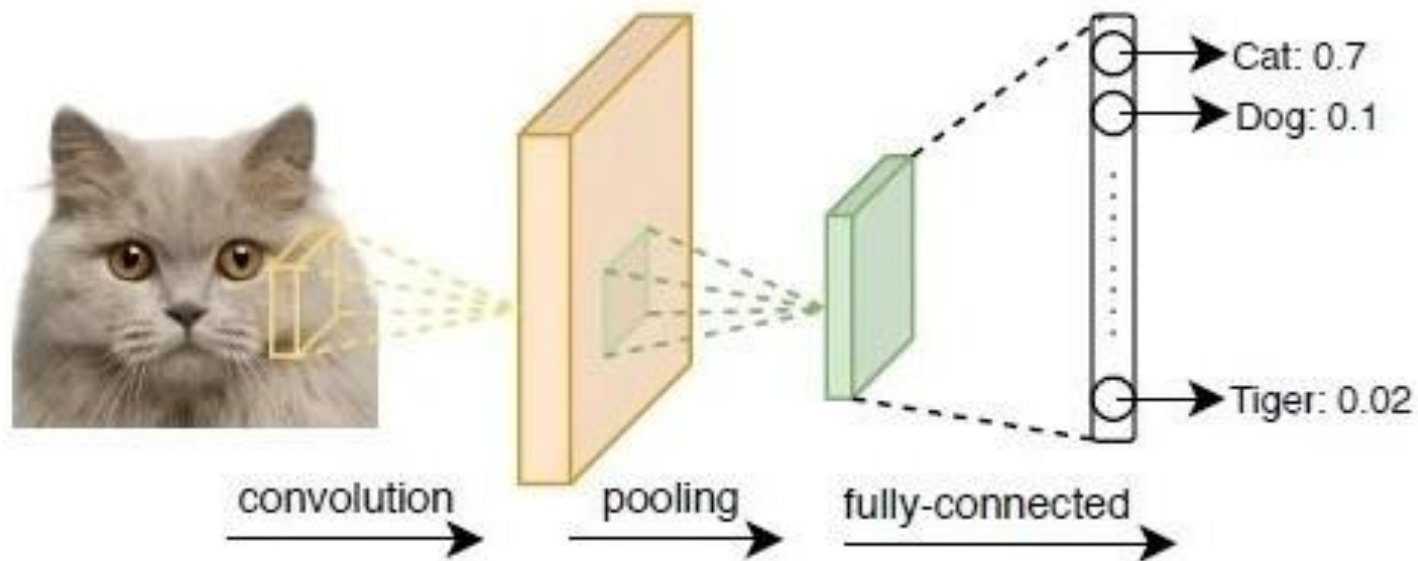
INPUT IMAGE



notebook



Convolutional Neural Network



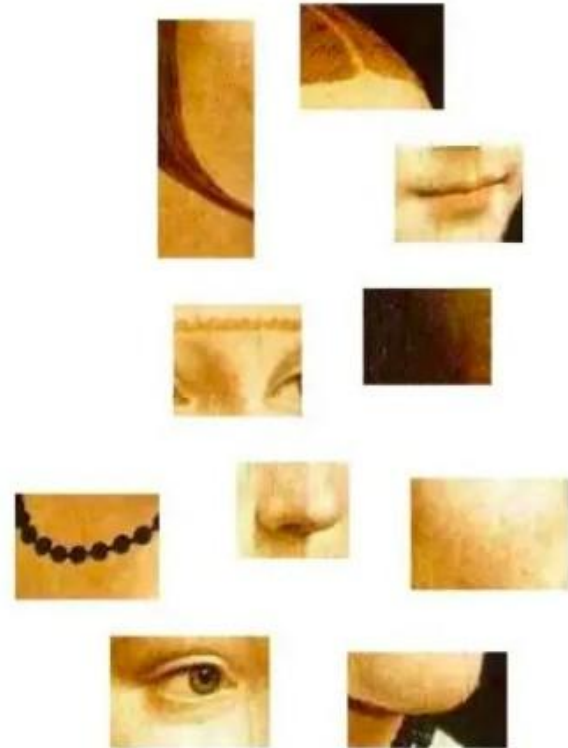
Bag of Features

- Bag of Features (BoF) is a technique used in computer vision and image processing to **extract and represent features** from images in a compact and meaningful way.
- The basic idea behind BoF is to **extract local features** from an image, such as **SIFT, SURF, or ORB**, and then **use clustering techniques** to group the features into a set of visual words.
- Each **image** is then **represented by a histogram** of these visual words, which is called **a bag of features**.

Bag of Features



Face



features

Bag of Features

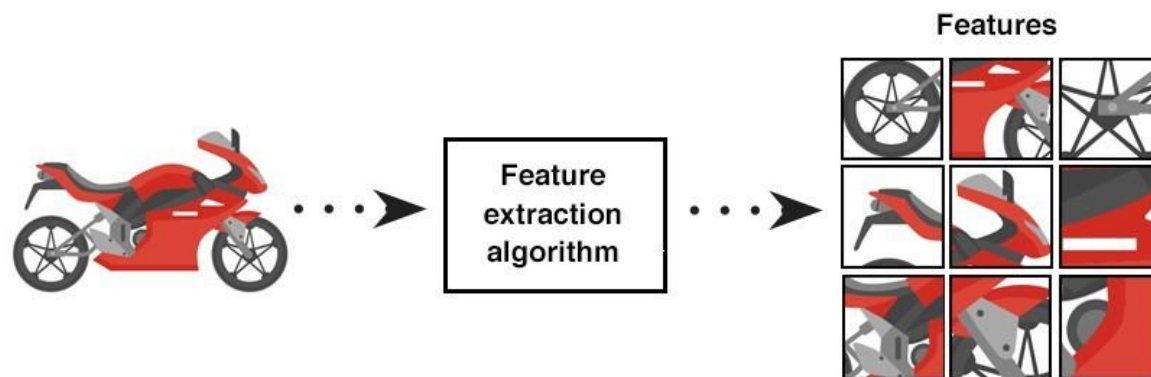
- The Bag of features representation is used in many computer vision and image processing tasks such as image retrieval, object recognition, and semantic segmentation.
- In image retrieval, BoF is used to represent images compactly and efficiently, allowing for fast and accurate retrieval of similar images.
- In object recognition, BoF extracts features from images and trains a classifier to recognize objects in new images.
- In semantic segmentation, BoF is used to extract features from images and train a model to predict the semantic labels of the pixels in the image.

Steps Involved in The BoF Process

- The basic steps involved in the Bag of Features (BoF) method include
 - feature extraction,
 - clustering, and
 - histogram representation.

Feature Extraction

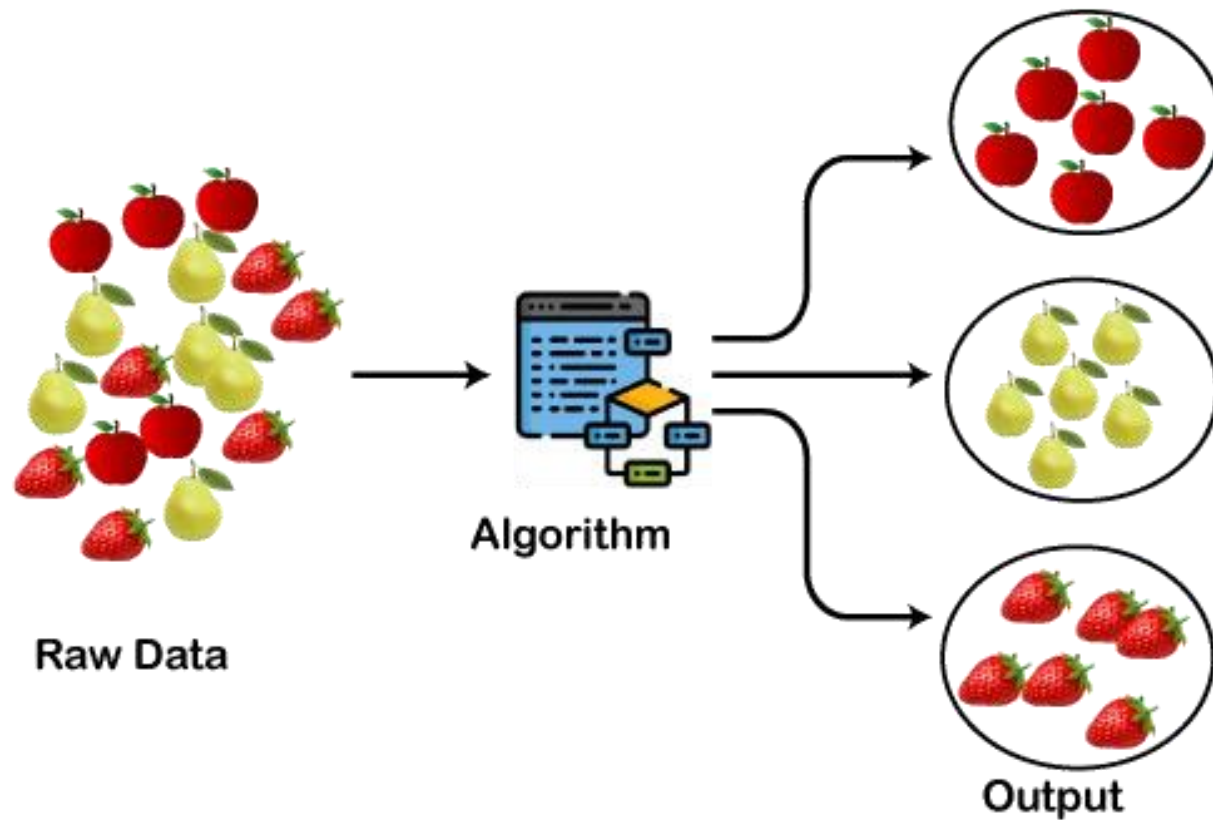
- The first step in the BoF method is to **extract local features from the images**.
- This is done using feature detection and description methods such as SIFT, SURF, or ORB. These methods **extract a set of key points and associated descriptor vectors** from the image.



Clustering

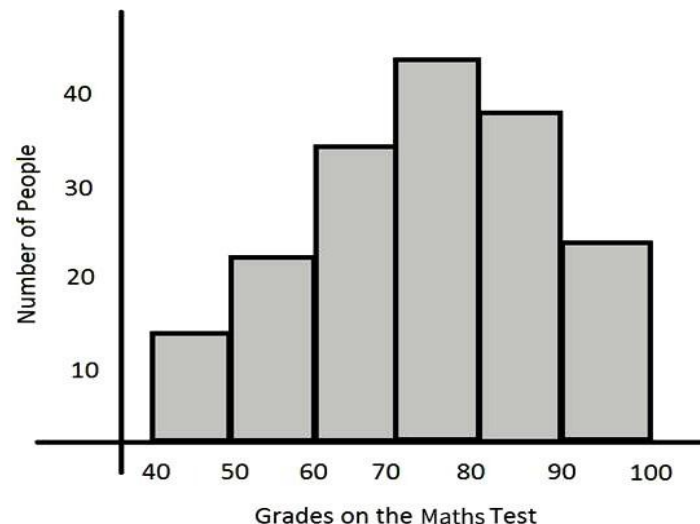
- The next step is to group the extracted features into a set of visual words.
- This is done by applying clustering techniques such as k-means or hierarchical clustering to the descriptor vectors.
- The result is a set of clusters, where each cluster represents a visual word.

Clustering



Histogram

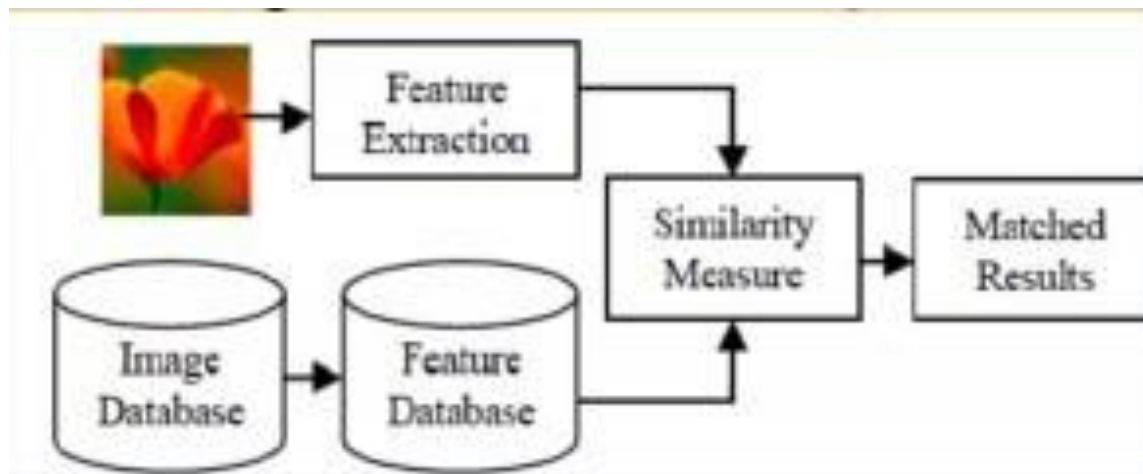
- Finally, the image is represented by a histogram of the visual words.
- This is done by **counting the number of features** that belong to each visual word and creating a histogram of these counts.
- The resulting histogram is the **bag of features representation** of the image.



Bag of Features: Real World Use Cases

- **Image retrieval:**
 - BoF has been used in image retrieval systems to represent images compactly and efficiently, allowing for **fast and accurate retrieval of similar images**.
 - It has been used in applications such as **image search engines**, where users can **search** for images **based on keywords or visual similarity**.

Bag of Features: Real World Use Cases



Bag of Features: Real World Use Cases

- **Object recognition:**
 - BoF has been used to **extract features** from images and **train a classifier** to recognize objects in new images.
 - It has been used in applications such as **surveillance systems**, where it can be used to recognize and track objects in video streams automatically.

Bag of Features: Real World Use Cases



Bag of Features: Real World Use Cases

- **Semantic segmentation:**
 - BoF has been used to extract features from images and train a model to **predict the semantic labels** of the pixels in the image.
 - It has been used in applications such as **autonomous driving**, where it can be used to **segment the road, vehicles, pedestrians, and other objects** in an image.

Bag of Features: Real World Use Cases



predict



Person
Bicycle
Background

Bag of Features: Real World Use Cases

- **Medical imaging:**
 - BoF has been used in medical imaging to segment and classify the structures in medical images such as CT and MRI scans.
 - It allows for the **automated detection** and **diagnosis of diseases**, making it a valuable tool for radiologists and physicians.
- **Robotics:**
 - BoF has been used in **robotics for object recognition and localization**.
 - It allows robots to **understand their environment** and **identify objects**, allowing them to interact with the environment more effectively.

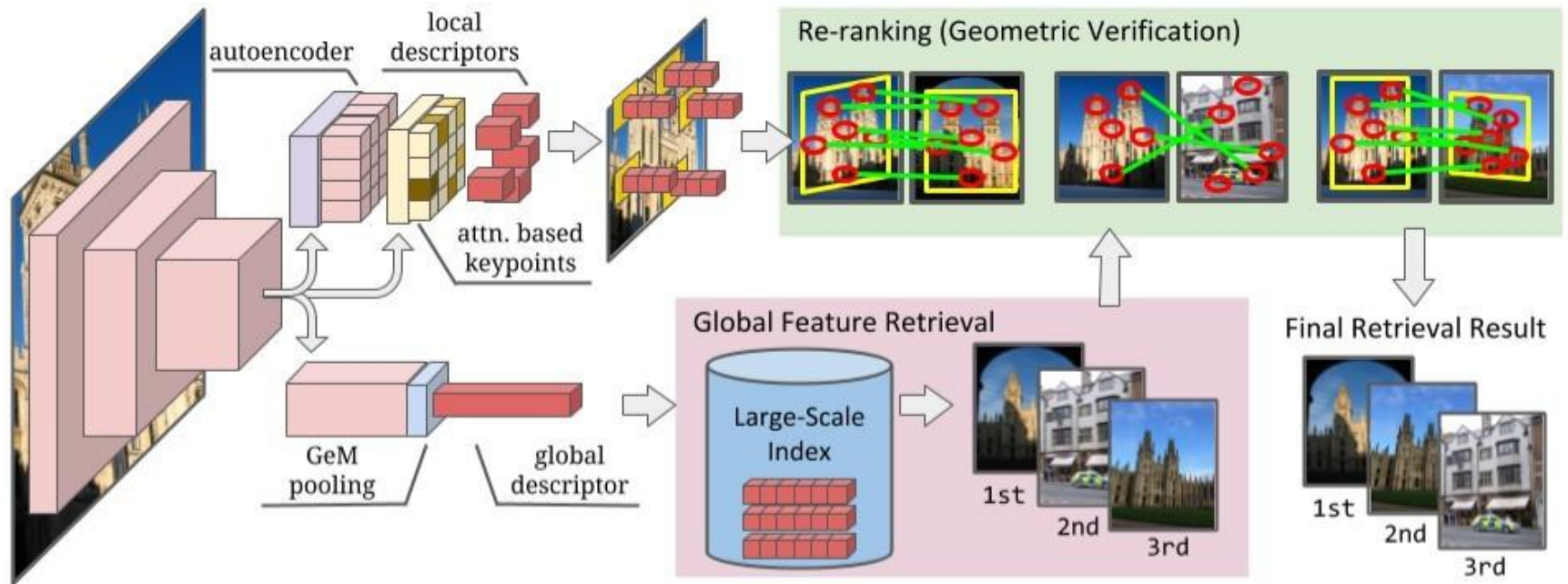
Large-scale Instance Recognition

- Large-scale instance recognition (LIR) is a computer vision task that aims to **recognize a specific instance of an object in a large dataset** of images.
- This is a challenging task because it **requires** the **model** to be able to identify the **same object** even when it appears in different poses, sizes, and illumination conditions.

Large-scale Instance Recognition

- LIR has a wide range of applications, such as:
 - **Image retrieval**: LIR can be used to **retrieve images of a specific object from a large dataset**. For example, you could use LIR to find all the images of the Eiffel Tower in a collection of millions of images.
 - **Object tracking**: LIR can be used to **track the movement of an object in a video**. This can be useful for applications such as surveillance or robotics.
 - **Scene understanding**: LIR can be used to **understand the context of an image**. For example, you could use LIR to determine that an image of a car is in a parking lot.

Large-scale Instance Recognition



Transfer Learning

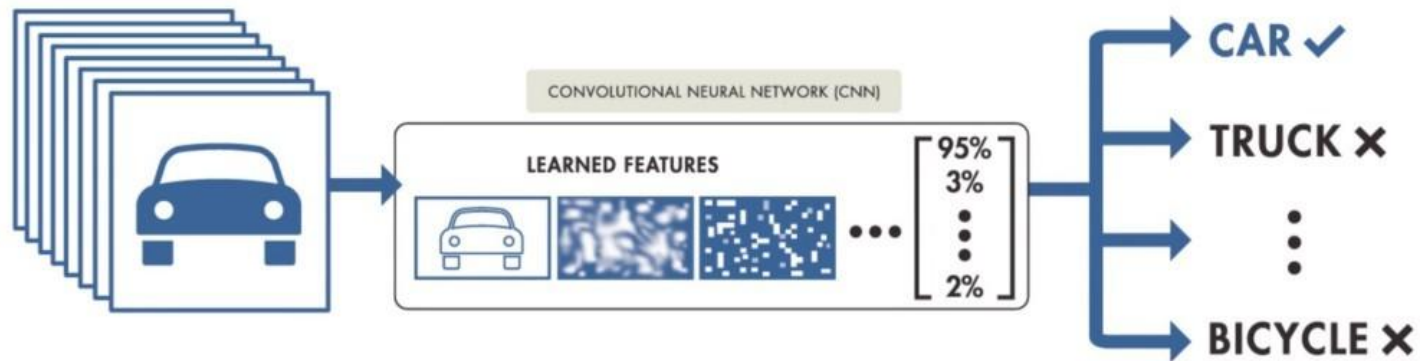
- Transfer learning is a machine learning technique where a **model trained on one task is reused as the starting point for a model on a second task.**
- This can be done by **freezing the weights of the first model** and then training the second model on a new dataset.
- This can help to **improve the performance of the second model**, as it can start from a point where the first model has already learned some important features.

Transfer Learning

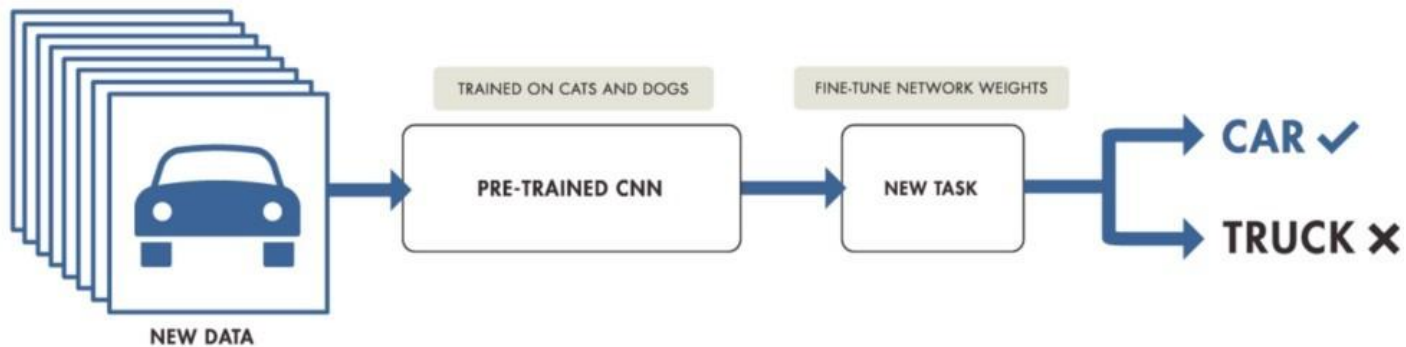
- Transfer learning is a popular approach in deep learning, as it can help **to reduce the amount of data** that needs to be collected for the second task.
- This can be especially useful when the second task has a small dataset, as it can be difficult to train a model from scratch on such a small dataset.

Transfer Learning

TRAINING FROM SCRATCH

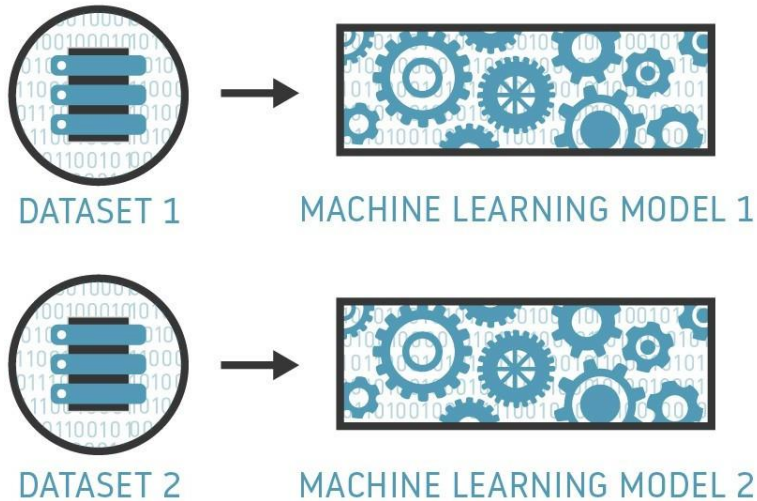


TRANSFER LEARNING

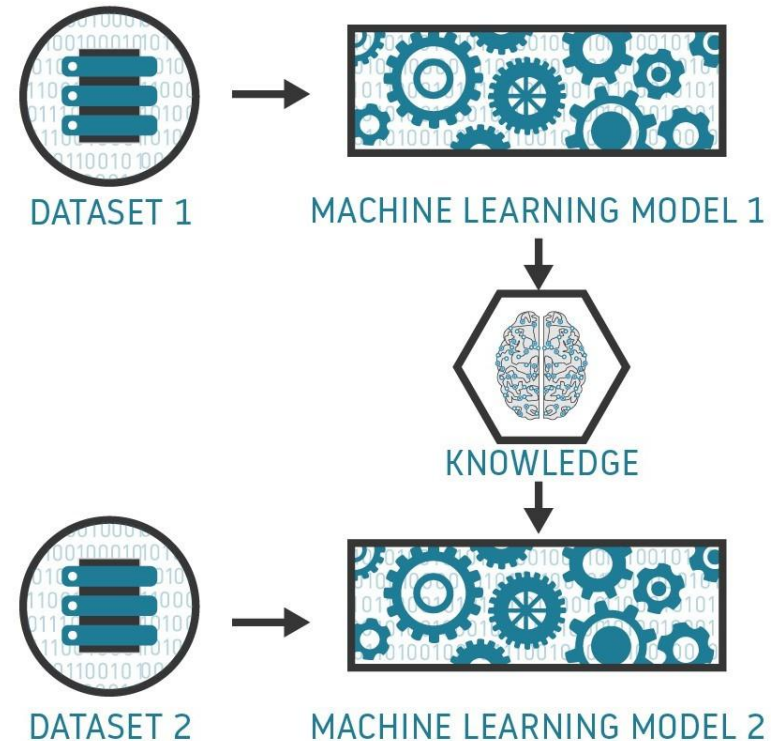


Transfer Learning

TRADITIONAL MACHINE LEARNING



TRANSFER LEARNING



Transfer Learning: Benefits

- **Reduces the amount of data needed:** Transfer learning can help to reduce the amount of data needed to train a model. This can be especially useful when the second task has a small dataset.
- **Improves performance:** Transfer learning can help to improve the performance of a model. This is because the first model has already learned some important features, which can be reused by the second model.
- **Saves time:** Transfer learning can save time, as it does not require the second model to be trained from scratch.

Transfer Learning: Limitations

- **The first model must be trained on a related task:** The first model must be trained on a task that is related to the second task. This is because the first model will learn features that are relevant to the second task.
- **The second model may not generalize well:** The second model may not generalize well to new data. This is because the second model is only trained on a subset of the possible data.
- **The first model may not be available:** The first model may not be available. This is because the first model may have been trained by someone else, or it may not be publicly available.

Transfer Learning: Develop Model Approach

- 1. Select Source Task.** You must select a related predictive modeling problem with an abundance of data where there is some relationship in the input data, output data, and/or concepts learned during the mapping from input to output data.
- 2. Develop Source Model.** Next, you must develop a skillful model for this first task. The model must be better than a naive model to ensure that some feature learning has been performed.

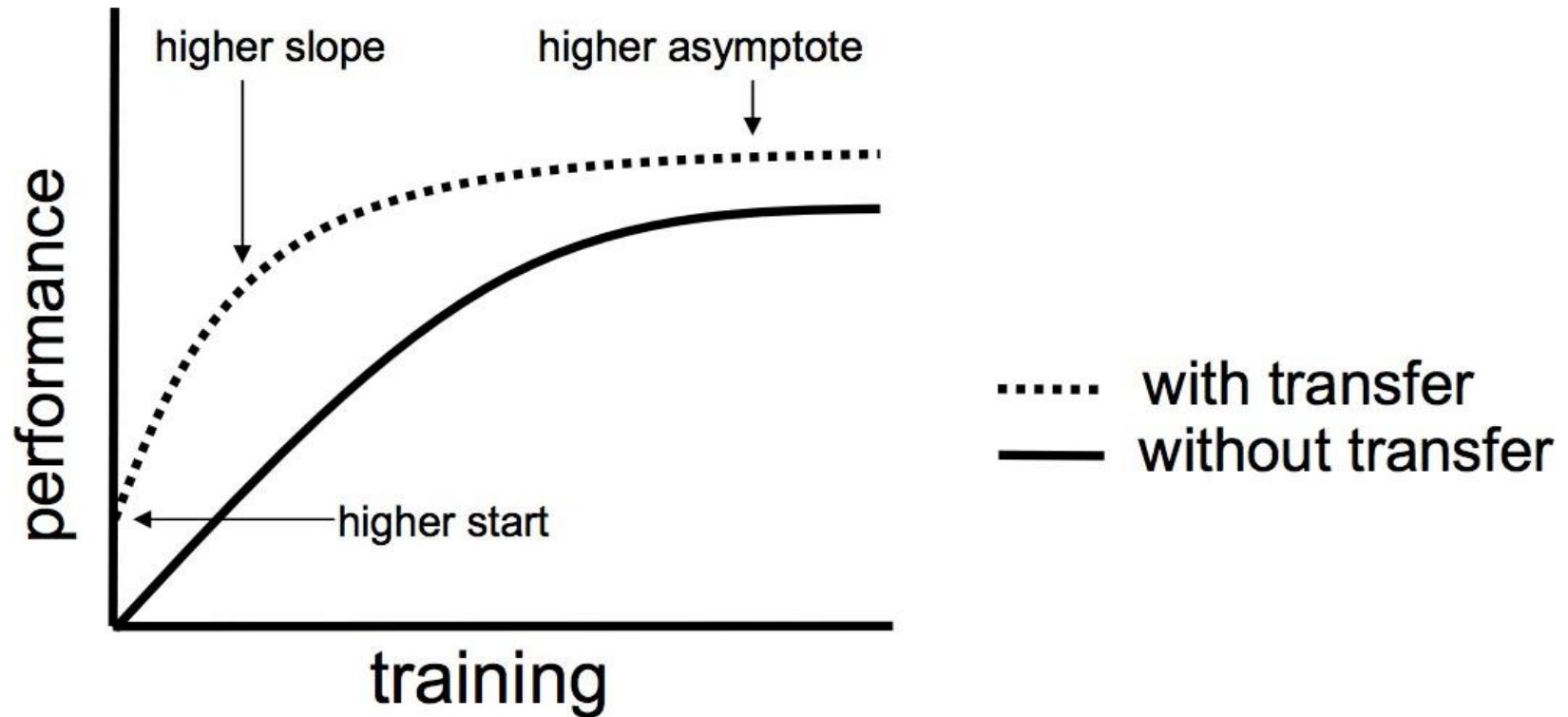
Transfer Learning: Develop Model Approach

3. **Reuse Model.** The model fit on the source task can then be used as the starting point for a model on the second task of interest. This may involve using all or parts of the model, depending on the modeling technique used.
4. **Tune Model.** Optionally, the model may need to be adapted or refined on the input-output paired data available for the task of interest.

Pre-trained Model Approach

- **Select Source Model.** A **pre-trained source model** is chosen from available models. Many research institutions release models on large and challenging datasets that may be included in the pool of candidate models from which to choose from.
- **Reuse Model.** The model **pre-trained model** can then be used as the starting point for a model on the second task of interest. This may involve using all or parts of the model, depending on the modeling technique used.
- **Tune Model.** Optionally, the model may need to be adapted or refined on the input-output pair data available for the task of interest.

Transfer Learning: Why?



- Higher start : better initial performance because useful knowledge has already been learned.
- Higher slope : the model improves its performance faster during training.
- Higher asymptote : better final performance after training has converged.

Notes

NumPy

- Numerical calculations and arrays

Pandas

- Pandas is a Python library used for working with data, especially data arranged in rows and columns, like an Excel spreadsheet.
- `pip install pandas`
- `dataframe=pd.read_csv('../../.csv')`

Matplotlib

- Creating graphs and plots

Seaborn

- Seaborn is a Python library used to create statistical graphs and visualizations.

END